

# AI Role

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FoodLoop's AI sits between the physical product and the digital marketplace. Its core job is to read product labels, extract expiry dates, classify products, and recommend discounts — reducing the manual work merchants have to do while keeping expired or misclassified items off the platform.

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## What the AI Does

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### 1. OCR — Read Product Labels (Sprint 2)

When a merchant uploads a product image, the AI pipeline:

1. Runs OCR on the image to extract all visible text from the label.
2. Passes the text to GPT Vision for structured extraction:
  - Product name
  - Expiry date
  - Category hint
3. Stores the result in `AIRecognitionResults`:
  - `DetectedProduct` — identified product name
  - `ExtractedText` — raw OCR output (kept for auditing)
  - `ExtractedExpiryDate` — parsed expiry date
  - `ConfidenceScore` — 0.0–1.0 indicating how certain the AI is

### 2. Product Classification (Sprint 2)

Based on the extracted text and image, the AI suggests which `Category` the product belongs to (Dairy, Bakery, Produce, etc.). This

pre-fills the category field in the product form, saving the merchant from picking manually.

### 3. Discount Calculation (Sprint 3)

The AI calculates a recommended `DiscountedPrice` for each listing using a rule-based pricing model that factors in:

- **Days until expiry** — the closer to the expiry date, the deeper the discount
- **Original price** — percentage-based calculation
- **Category norms** — some categories (e.g. fresh produce) depreciate faster than others
- **Historical sell-through rates** — if a product type consistently sells slowly, the AI recommends an earlier and deeper discount

The formula follows a decay curve: a product 7 days from expiry might get 20% off, while a product 1 day from expiry gets 60%+ off.

### 4. Scheduled Repricing (Sprint 4)

A background job runs daily to re-evaluate all active listings. As `ExpirationDate` approaches, prices are adjusted downward automatically according to the decay curve. This keeps pricing accurate without merchants having to update listings manually.

### 5. Performance Monitoring (Sprint 5)

The AI tracks its own accuracy:

- Compares `ExtractedExpiryDate` against the date the merchant confirmed
  - Tracks how often its category suggestions are accepted vs. corrected
  - Surfaces metrics in the admin analytics dashboard for team review and model improvement
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# What the AI Can Update Autonomously

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These actions happen without human intervention when `ConfidenceScore` is above a defined threshold:

| Action                                          | Trigger                 | Condition                                    |
|-------------------------------------------------|-------------------------|----------------------------------------------|
| Set <code>ProductListing.ExpirationDate</code>  | After OCR               | <code>ConfidenceScore &gt;= threshold</code> |
| Set <code>ProductListing.DiscountedPrice</code> | After expiry extraction | Expiry date is confirmed                     |
| Reprice an existing listing                     | Daily background job    | Listing is still active                      |
| Pre-fill category suggestion                    | After classification    | Always (suggestion, not forced)              |

The confidence threshold is configurable. In early sprints it is set conservatively so more items go through human review while the model is being evaluated.

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# What Requires Human Review

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These actions are queued for review when confidence is below the threshold, or always by policy:

| Action                                                                    | Who reviews | Screen                                                          |
|---------------------------------------------------------------------------|-------------|-----------------------------------------------------------------|
| Low-confidence OCR result ( <code>ConfidenceScore &lt; threshold</code> ) | Admin       | Moderation queue ( <code>moderation_queue_active_state</code> ) |

|                                          |          |                                                                |
|------------------------------------------|----------|----------------------------------------------------------------|
| Expiry date could not be extracted       | Merchant | Product creation form (manual entry)                           |
| Category suggestion rejected by merchant | Merchant | Product form                                                   |
| Product flagged by a user report         | Admin    | Moderation queue                                               |
| Duplicate listing detection              | Admin    | Moderation queue                                               |
| Missing or expired health permit         | Admin    | Moderation queue                                               |
| Store document verification              | Admin    | GET /admin/stores/pending →<br>PATCH /admin/stores/{id}/verify |

The moderation queue shows the AI's confidence score on each card so the admin can triage quickly: high-confidence items get approved in seconds, low-confidence items get careful review.

## The AIRecognitionResult Record

Every OCR job produces one AIRecognitionResult linked to a ProductListing (1-to-1):

|                     |                                      |
|---------------------|--------------------------------------|
| AIRecognitionResult |                                      |
| ListingId           | → the product this result belongs to |
| DetectedProduct     | → "Labneh 500g" (from label)         |
| ExtractedText       | → full raw OCR output                |
| ExtractedExpiryDate | → 2026-08-14 (parsed from "Best Bef  |
| ConfidenceScore     | → 0.87                               |
| Reviewed            | → false (until an admin or merchant  |

When Reviewed = true it means a human has confirmed the AI's

output and it can be trusted for pricing calculations.

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## Data Flow: Image → Listed Product

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```
Merchant uploads photo
      ↓
AI Pipeline (OCR + GPT Vision)
      ↓
AIRecognitionResult stored
      ↓
ConfidenceScore >= threshold?
├── YES → ExpirationDate auto-filled
│       DiscountedPrice calculated
│       Listing published
└── NO  → Queued in moderation
        Admin reviews
        Admin approves/rejects
              ↓
        Listing published (or rejected)
```

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## Pricing Formula (Rule-Based, Sprint 3)

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The discount is a function of days remaining until expiry:

```
daysLeft = ExpirationDate - today

if daysLeft > 7:
    discount = 10%
elif daysLeft > 3:
    discount = 25%
elif daysLeft > 1:
    discount = 45%
```

```
else:
    discount = 60%+

DiscountedPrice = OriginalPrice * (1 - discount)
```

This is the initial rule-based version. Sprint 4 replaces it with a scheduled repricing job that re-evaluates every listing daily and applies the decay curve progressively rather than in fixed steps.

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## What the AI Does NOT Do (boundaries)

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- It does not directly place or cancel orders.
- It does not directly approve or reject stores — that is always admin-only.
- It does not set a price higher than `OriginalPrice`.
- It does not delete listings — it can flag them, but removal requires human action.
- It does not access user personal data for pricing decisions.